Impact of UAV Motion on Free-Space Optical Link Stability and Receiver Design Optimization

Saadman Yasar, Zaheen E. Muktadi Syed, Md Sarwar Uddin Chowdhury, Sk. Samiul Reza, Murat Yuksel

***Abstract*—Free-space optical (FSO) communication using unmanned aerial vehicles (UAVs) offers a high-capacity solution for airborne data links, but maintaining beam alignment remains a significant challenge due to vibration-induced motion. In UAVs, the vibration originates from high-frequency oscillations caused by spinning rotors and structural resonance, while hover-induced motion arises from low-frequency drift and attitude fluctuations resulting from control system dynamics. In this study, we perform controlled hovering flights with a quadrotor UAV to measure deviations in position, orientation, linear velocity, and angular velocity. Using the collected data, we calculate pointing errors and assess their impact on the FSO link stability between two UAVs. To analyze how UAV motion dynamics affect optical signal propagation, we simulate the FSO link in Ansys Zemax OpticStudio (ZOS) using ray-tracing techniques. The simulations model how pointing errors influence received power under atmospheric attenuation and turbulence. We explore the relationship between received optical power and receiver aperture area under both stationary and vibrating conditions. Based on the results, we explore how the shape and size of the receiver affect alignment tolerance. Our findings provide practical guidelines for optimizing receiver aperture dimensions to improve the robustness of FSO communication systems for UAVs. Complementary hexacopter flight data are also analyzed to validate the generality of the observed trends across UAV platforms.**

I. Introduction

Free-space optical (FSO) communication is emerging as a high-capacity solution for line-of-sight (LoS) wireless links, offering advantages such as unregulated spectrum, high data rates, and resilience to electromagnetic interference [1]. With the growing demand for rapid and flexible deployment of communication infrastructure, the integration of FSO with unmanned aerial vehicles (UAVs) has drawn substantial research attention. UAV-mounted FSO terminals can provide dynamic connectivity in infrastructure-deficient or remote environments, enabling applications such as aerial backhaul/fronthaul, disaster recovery, and tactical communication [2]–[4].

Despite its potential, the deployment of UAV-based FSO systems presents significant technical challenges. One of the most critical issues is maintaining accurate beam alignment between the transmitter and receiver. Unlike fixed ground terminals, UAVs are subject to motion and vibration that introduce pointing errors, reducing received optical power and link reliability [5]–[7]. These errors can be broadly categorized into high-frequency and low-frequency regimes. High-frequency oscillations (typically 40–100 Hz) are primarily caused by rotor spin and structural resonance, resulting in angular deviation that destabilizes the beam footprint [5], [8]. Conversely, low-frequency hover-induced motion (typically <1 Hz) arises from environmental disturbances (e.g. wind) and control loop dynamics, leading to gradual drift in position and orientation [2], [6].

Attaining robust FSO communication links among flying platforms depends heavily on understanding the impact of the environmental and operational factors on the optical beam alignment. Recent studies have proposed analytical models for characterizing pointing error distributions [6], [8], while others have investigated optimized transceiver configurations to counteract mobility and misalignment [3]. To quantify these mobility-induced effects, a previous experimental study investigated the hover state dynamics of a quadrotor UAV by measuring vibration-driven and hover-induced deviations across 12 dynamic states, including position, orientation, linear velocity, and angular velocity, on the x, y, and z axes [8]. Displacements, disorientations, and their rate of change were estimated from two distinct hover flights to formulate a measurement-driven model of motion-induced FSO beam misalignment. However, there remains a gap in integrating experimental flight data with optical ray-tracing simulations to predict performance under real-world UAV motion.

In this work, the key contributions are:

- A systematic framework that facilitates studying the influence of real motion data on FSO link performance. The framework enables injection of real motion data to Zemax OpticsStudio, known for its accuracy in optics simulations, and observe how such motion influences the FSO link performance.
- Controlled flights of a quadrotor UAV to collect fine-grained motion telemetry used as input to the optical link simulation framework.
- Leveraging of publicly available hexacopter flight datasets to conduct experiments and simulations, enabling a consistent comparison across UAV platforms.
- Exploration of how receiver aperture shape and dimensions influence beam coupling efficiency under dynamic

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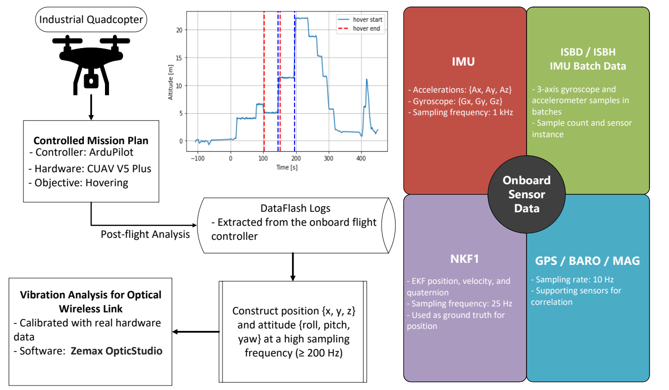

Fig. 1. A systematic framework for studying vibration effect on FSO communication.

UAV conditions.

- Design recommendations for robust FSO link configurations, optimal receiver aperture configuration in UAV-based FSO networks.

The remainder of this paper is organized as follows: In section II, we discuss previous work related to UAV vibration analysis. Section III presents the experimental UAV setup and data acquisition methodology. Section IV describes the pointing error model and the ZOS simulation framework to explore coupling efficiency. Section V provides a study on receiver area optimization. Finally, Section VI concludes the paper and outlines future directions.

II. Related Work

UAV-induced motion and vibration pose significant challenges to maintaining precise beam alignment in FSO communication systems. Prior studies have explored various aspects of this problem. Pottoo et al. [8] have conducted controlled hover flights and developed a measurement-based model to characterize UAV motion dynamics, offering valuable insights into position and orientation fluctuations. However, their work does not assess how such dynamics affect optical beam propagation or coupling efficiency. Wang et al. [9] present an analytical framework for UAV-based FSO links, incorporating turbulence, pointing error, and angle-of-arrival (AOA) fluctuations, and derive outage probability expressions, but their model does not incorporate experimental flight data.

Other efforts have focused on enhancing FSO link performance in non-terrestrial platforms. Ata and Alouini [10] have applied adaptive optics to mitigate turbulence in High-altitude platform station (HAPS)-based links, while Yahia et al. [11] and Deka et al. [12] have explored hybrid (radio frequency) RF/FSO architectures with performance optimization under various link conditions. Zedini et al. [13] and Singh et al. [14] have introduced multiple-input and multiple-output (MIMO)-based models for improving link robustness and diversity in HAPS scenarios. While these works provide valuable theoretical and analytical foundations, they do not explicitly examine the impact of real UAV mobility on optical link behavior or receiver design.

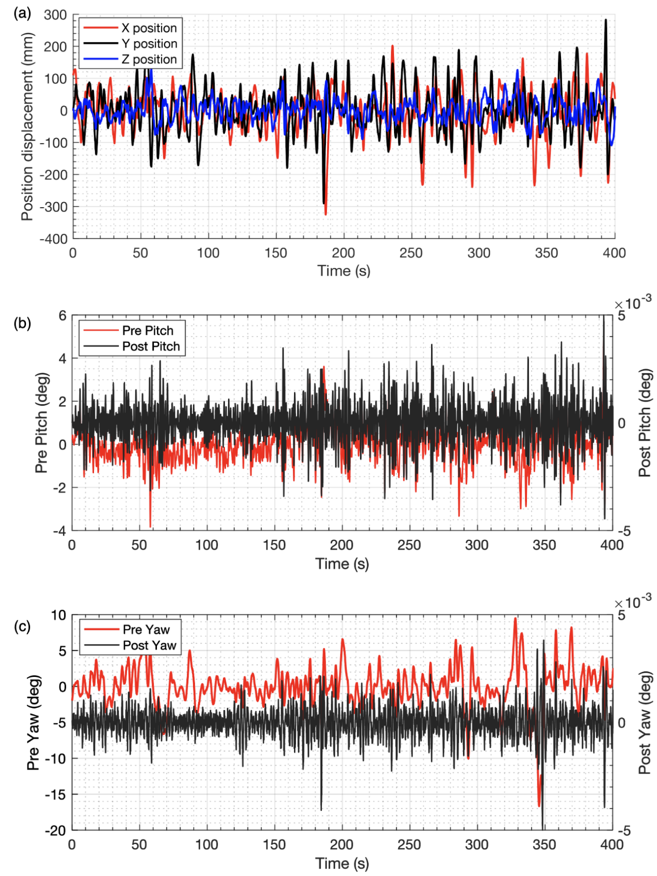

Fig. 2. (a) Measured UAV translational displacement along the x, y, and z axes during the hovering interval. (b,c) Measured pre-gimbal angle and modeled post-gimbal residual jitter for pitch and yaw.

In this work, we develop a measurement-informed ZOS-based simulation framework for UAV-mounted FSO links. Unlike prior telemetry-assisted studies, this work integrates realistic UAV hover motion with detailed optical propagation, including post-gimbal residual jitter, atmospheric absorption, Mie scattering, Kolmogorov phase-screen scintillation, receiver optics, and receiver-to-detector optical flow. The framework is used to quantify the effects of UAV vibration and hover drift on beam alignment, received power, coupling efficiency, and communication performance, while specifically addressing receiver-area and receiver-shape optimization for robust UAV-FSO link design.

III. Empirical Framework

The experimental workflow we use to evaluate the impact of UAV-induced vibrations on an FSO link during hovering is illustrated in Fig. 1. We operate an industrial quadcopter (Aris M1200) equipped with a CUAV V5 Plus flight controller running ArduPilot firmware [15], [16] to perform a controlled hovering mission. During flight, we log onboard sensor data, including high-frequency raw inertial measurement unit (IMU) measurements, where accelerometer and gyroscope information is reported through separate messages at up to 1000 Hz, batch-mode IMU data (ISBD/ISBH), and low-frequency GPS, barometer, and magnetometer readings at 10 Hz. We also

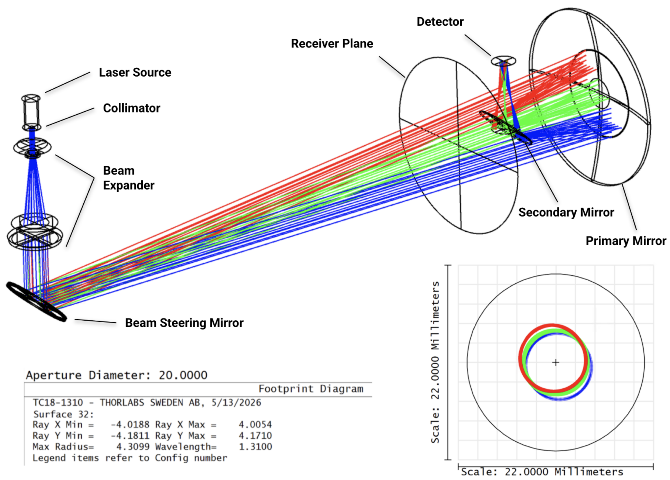

Fig. 3. ZOS model of the FSO system. The left panel shows a ray-trace diagram with labeled components including the laser source, collimator, beam expander, steering mirror, and receiver plane. Three beam footprints on the receiver, shown on the right, represent different configurations simulating angular misalignments due to drone vibration.

record Extended Kalman Filter (EKF) outputs (NKF1), which provide position, velocity, and orientation at 25 Hz and are used to validate the reconstructed motion trends.

After the flight, the flash logs are extracted, time-aligned, and interpolated onto a common time base to reconstruct the UAV position and attitude sequence $\{x, y, z, \text{roll}, \text{pitch}, \text{yaw}\}$ at 400 Hz.

We identify the hovering interval from the altitude profile and use the reconstructed motion data in ZOS to perform vibration analysis. In addition to the quadcopter-based experiments, we follow the same processing and analysis pipeline using flight data collected from a hexacopter platform provided by the Aerial Experimentation and Research Platform for Advanced Wireless (AERPAW) testbed [17]. AERPAW provides high-fidelity flight logs for a six-rotor UAV operating under controlled hovering conditions, enabling a direct comparison between quadcopter and hexacopter motion characteristics. This workflow enables us to calibrate and assess the stability of the FSO link under realistic in-flight conditions.

Fig. 2 illustrates the quadcopter's recorded motion dynamics during flight. Subplot (a) shows the translational positions (x, y, z) and Subplot (b,c) shows the angular orientation in terms of pitch and yaw over time. These fine-scale translational and rotational disturbances are later used in beam alignment simulations to assess robustness under realistic UAV conditions. Both the quadcopter (Aris M1200) and hexacopter (AERPAW) motion datasets used in this study are made publicly available for reproducibility and further analysis and can be accessed here [18].

A. UAV-Compatible Optical System Design

To simulate the optical communication link under realistic drone-induced perturbations, a detailed ray-tracing model was developed in ZOS 2022 R2.02 [19]. ZOS was selected for this study due to its ability to simulate physical ray behavior with high fidelity. It offers precise modeling of beam divergence,

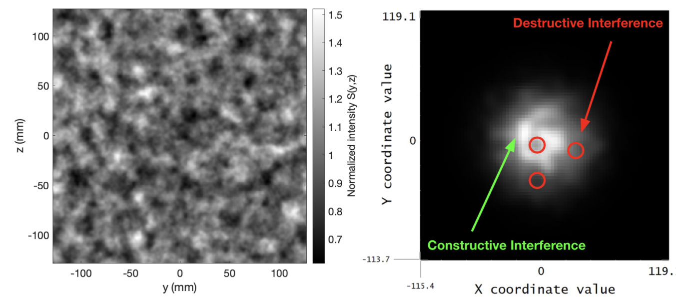

Fig. 4. Scintillation modeling using cumulative Kolmogorov phase-screens and the corresponding irradiance distribution in the detector.

wavelength-dependent propagation, optical apertures, and real-world alignment errors. Unlike simplified analytical tools, ZOS enables us to import real hardware parameters (e.g., lens prescriptions, mounting offsets) and study the full propagation path from source to receiver. This physical accuracy, combined with programmable interfaces like the ZOS application programming interface (API), allows us to create a scalable and automated simulation framework that directly integrates drone telemetry into optical alignment evaluation. The optical system is constructed using commercially available components from Thorlabs [20], [21], including collimators and beam expanders, to ensure reproducibility and experimental relevance.

Atmospheric channel effects are incorporated in ZOS by modeling absorption and scattering deterministically, while scintillation is treated stochastically. Absorption is simulated by assigning an extinction coefficient through the complex refractive index of the material filling the propagation volume, from which the absorption coefficient can be related to Beer–Lambert attenuation over the link distance. Scattering is implemented using the Mie-scattering model in the non-sequential volume, with the link distance, particle refractive index, particle radius, and mean free path defining the scattering strength. In contrast, scintillation is modeled using multiple randomly generated Kolmogorov phase screens placed at equal intervals between the transmitter and receiver. These screens introduce random phase perturbations due to atmospheric refractive-index fluctuations, producing constructive and destructive interference patterns in the detector irradiance distribution. Fig. 4 verifies the scintillation implementation by showing the cumulative Kolmogorov phase-screen intensity variation and the corresponding Zemax detector irradiance pattern, where random phase perturbations produce localized constructive and destructive interference across the received beam footprint. This implementation highlights the advantage of ZOS for visualizing how atmospheric loss and turbulence modify the received optical footprint, irradiance distribution, and collected link power. The specific ZOS settings and simulation parameters used for these atmospheric channel models are provided in the accompanying simulation files [22].

Fig. 3 illustrates the ZOS layout, which consists of a fiber-coupled laser source, a collimator, a beam expander, and a beam steering mirror mounted at 45°, directing the beam toward a distant receiver plane. The beam is then collected by a focusing optical setup consisting of a primary mirror and a secondary mirror, which concentrates the received optical beam onto the detector. The bottom-right plot shows the beam footprint on the receiver for three configurations, each representing distinct angular misalignments induced by drone vibrations. These configurations are handled by using ZOS' multi-configuration feature, allowing batch simulation of different tilt/shift scenarios without manual reconfiguration.

Beam steering and vibration effects are emulated in ZOS using a sequence of Coordinate Break surfaces placed along the transmitter and receiver optical path. These Coordinate Breaks allow controlled changes in beam direction and receiver alignment by introducing translational and angular perturbations into the optical system. Translational shifts are applied by varying ZOS' Variables and Parameters associated with these Coordinate Break surfaces, enabling dynamic adjustment of the beam position. These adjustments are automated via the ZOS API, which enables batch ray-tracing simulations across a database of UAV vibration profiles derived from logged flight telemetry. The Zemax simulation files used in this study are available in [22].

To address the effect of angular jitter, the simulation includes post-gimbal residual pointing error. As illustrated in Fig. 2(b,c) the raw measured pitch and yaw signals contain both slow platform attitude variations and high-frequency angular vibration. In practice, a three-axis gimbal compensates the dominant low-frequency rotational motion, but a small residual angular vibration remains. The DJI Zenmuse X7 reports an angular vibration suppression accuracy of approximately $\pm 0.005^\circ$ [23]. Accordingly, the slow angular drift is removed from the measured pitch and yaw data, and the remaining high-frequency component is scaled to a bounded residual range. The resulting post-gimbal pitch and yaw residues are then used as angular perturbations in the ZOS model.

The use of a beam expander in the optical path plays a critical role in reducing beam divergence and improving spatial coherence at long distances. By increasing the beam diameter post-collimation, the expander effectively lowers the angular spread due to diffraction, which is essential for maintaining alignment in FSO links over several meters or more. The beam divergence was calculated by measuring the beam footprint size at the receiver plane across each configuration.

The ZOS simulations are used to generate the received beam footprint, centroid position, and spatial distribution on the receiver plane under UAV-induced translational and angular perturbations. Receiver area and shape are not implemented as a separate physical aperture stop in ZOS; instead, they are evaluated in post-processing. For each simulated realization, coupling is determined by checking whether the beam centroid falls within the prescribed receiver geometry: inside the selected radius for circular receivers, or within the selected width and height for rectangular receivers. The resulting centroid-hit statistics are then used to compute the coupling efficiency for different receiver areas and shapes under dynamic misalignment.

The proposed optical system is optimized for UAV de-

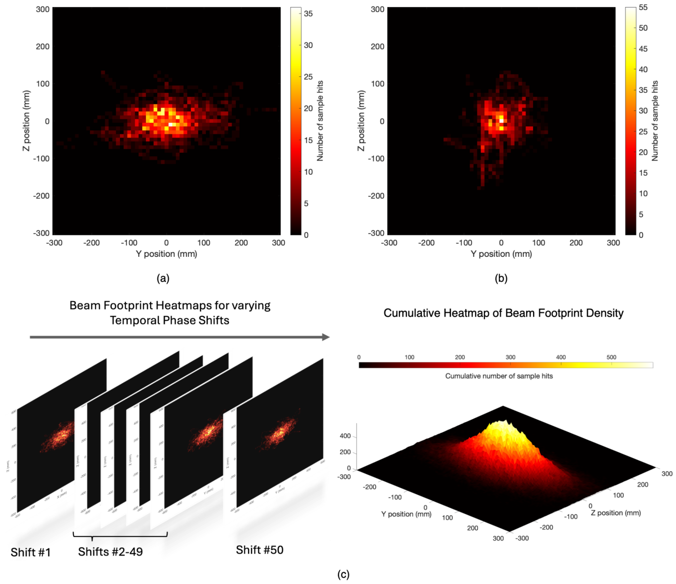

Fig. 5. Heatmap of beam-footprint density on stationary receiver at a 100 m link distance for (a) quadcopter and (b) hexacopter, where color denotes the frequency of beam landings within each spatial bin and (c) beam footprint evolution under receiver motion. (left) A sequence of 2D heatmaps corresponding to 50 temporal phase shifts applied to the receiver vibration data. (right) cumulative 3D heatmap showing the total beam footprint density across all shifts, illustrating the full spatial spread.

ployment based on size, weight, and power (SWaP) considerations. It comprises a Triplet Fiber Optic Collimator (TC18APC-1550) [20], a 5X Achromatic Galilean Beam Expander (GBE05-C) [21], and a 1550 nm optical transceiver. The triplet collimator, with a compact focal length of 18.28 mm and precision alignment via FC/APC coupling, ensures high-quality collimation with minimal divergence while maintaining a lightweight and compact form factor ($\approx$ 0.12 lbs). The Galilean beam expander further reduces beam divergence enabling improved long-range propagation without introducing significant beam divergence or bulk—ideal for weight-constrained platforms. Together, this setup minimizes optical aberrations and maintains alignment stability under dynamic conditions. The 1550 nm optical transceiver, commonly selected for its low atmospheric attenuation and compatibility with telecom-grade optics, ensures efficient power usage and reliable transmission. The overall size of the integrated optical stack, including mechanical housing and alignment tolerances, is estimated to be approximately 200 mm $\times$ 100 mm $\times$ 100 mm. This system architecture offers a high-performance yet low-footprint solution, making it particularly well suited for UAV-based FSO links where payload constraints and vibration tolerance are critical.

B. UAV to Stationary Receiver

To assess beam stability during drone-based FSO transmission, we analyze the beam landing distributions on a stationary receiver positioned 100 m from a hovering transmitter UAV for both quadcopter and hexacopter platforms. Fig. 5(a,b) present 2D heatmaps of beam-footprint density over the receiver plane, where each spatial bin indicates the frequency of beam-footprint incidence. The transmitter was held in hover mode for a sustained duration, and the receiver remained fixed and pre-aligned with the nominal beam axis. Beam centers were collected over time and discretized into a grid across the Y–Z plane. The resulting heatmap spans roughly 0.3 meters in both axes, with color intensity indicating the number of beam hits per spatial bin. Brighter regions correspond to areas of concentrated beam activity. Despite the UAV's stable hover, the beam footprints demonstrate notable spatial spread, with fluctuations driven primarily by translational micro-motions. The elongated distribution indicates lateral drift and vibrational excursions, highlighting the need for robust receiver aperture design or adaptive tracking mechanisms to ensure reliable FSO link performance.

C. UAV to UAV

To model real-world drone-based optical reception, we extend our analysis to include a moving receiver that experiences time-varying displacements due to UAV vibrations. In this setup, both the transmitter and receiver are drone-mounted, both exhibiting translational and angular motion recorded from UAV telemetry logs. Fig. 5(c) illustrates the heatmap generation pipeline for this dynamic case. Each individual heatmap (left) represents the spatial beam footprint at the receiver plane for a given temporal phase shift, corresponding to a delay in the synchronization between transmitter and receiver telemetry log. A total of 50 temporal shift scenarios are simulated using the ZOS API by applying different offsets to the receiver positional log. For the UAV-to-UAV case, the same measured flight telemetry is phase-shifted to generate diverse relative transmitter–receiver motion realizations while preserving the original UAV vibration statistics. Since link performance depends on the instantaneous relative alignment, these shifts provide physically plausible alignment scenarios; finer shift resolution does not appreciably change the coupling or receiver-area trends.

For each temporal shift, ZOS was used to ray trace the beam footprint at the receiver surface, producing a 2D histogram of beam center positions in the Y–Z plane. These 50 individual distributions were then cumulatively aggregated to construct a global heatmap (shown on the right), capturing the full envelope of misalignment over time. The final result is a high-resolution spatial density map that reflects the probability distribution of the beam center under drone motion. This quantifies the alignment challenge and informs optimal receiver sizing or the need for closed-loop tracking systems. The entire pipeline was automated using the ZOS API with MATLAB.

IV. Coupling Efficiency

The dependence of coupling efficiency—defined as the fraction of transmitted optical power captured by the receiver—on receiver area is evaluated for both quadcopter and hexacopter platforms under stationary and motion-induced conditions. Figs. 6(a) and (b) present coupling efficiency versus receiver area for quadcopters and hexacopters, respectively, considering rectangular apertures with varying aspect ratios (AR) as well

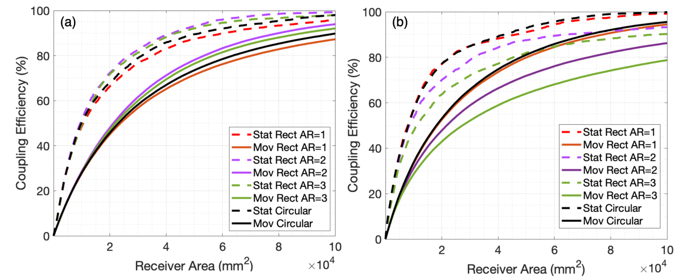

Fig. 6. Coupling efficiency as a function of receiver area for different aperture shapes and aspect ratios (a) for quadcopter and (b) for hexacopter platforms, respectively, comparing stationary and moving beam conditions for rectangular apertures (AR = 1–3) and circular apertures.

as circular apertures. Solid curves correspond to the moving case, where the receiver experiences experimentally measured platform motion, while dashed curves represent the stationary reference case. The hexacopter exhibits higher stability due to its increased rotor count, resulting in reduced platform-induced beam jitter.

A. Impact of Receiver Aperture Shape

Rectangular apertures with higher ARs (e.g., AR = 2 or 3) consistently outperform lower AR designs for the same total aperture area, particularly under stationary conditions. This advantage arises from their ability to capture beam wander more effectively along the axis of dominant vibration, i.e., the azimuth axis as seen from Fig. 6. In practice, the high-aspect-ratio receiver can be implemented using a rectangular slit aperture followed by receiver collection optics before the photodetector, allowing the effective collection geometry to be shaped without requiring a physically rectangular high-speed detector. For moving receivers, coupling efficiency increases more gradually, but high-AR designs still provide meaningful improvements by accommodating lateral deviation. Circular apertures (black curves) show a balanced response but lack the directional adaptability of elongated rectangular shapes. While they eventually reach near-total coupling efficiency, their performance is surpassed by rectangular designs in the moderate area regime. Overall, these results underscore the importance of receiver geometry in maximizing optical power capture.

To better understand the spatial characteristics of beam drift in a UAV-to-UAV FSO system, we analyze the misalignment

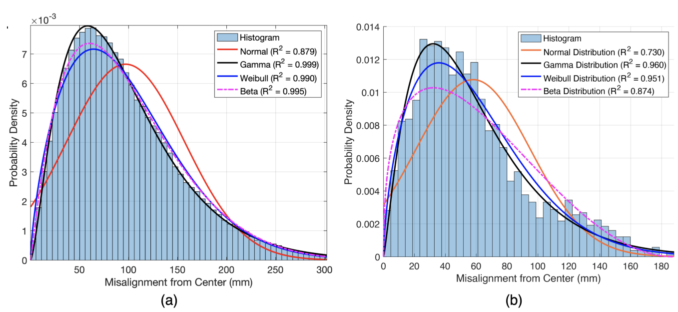

Fig. 7. Histogram of radial beam misalignment from the nominal receiver center under UAV motion: (a) quadcopter and (b) hexacopter, showing empirical distributions with fitted Normal, Gamma, Weibull, and Beta models and corresponding R^2 values.

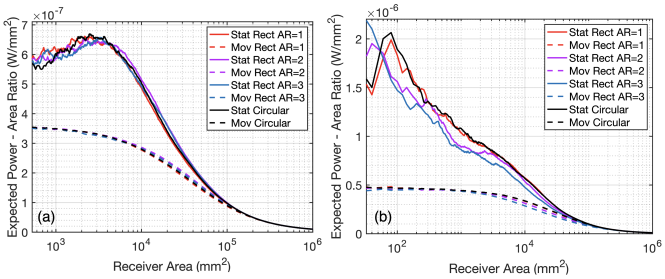

Fig. 8. Expected received power-to-area ratio versus receiver area in (a) quadcopter platform and (b) hexacopter platform for various receiver geometries and motion states.

of beam centers from the nominal optical axis under realistic UAV motion. Fig. 7 presents a statistical overview of the beam deviation across more than 100,000 samples simulated in ZOS under various temporal phase shifts and receiver positions. Fig. 7(a) and (b) show the distributions of radial misalignment from the nominal center for the quadcopter and hexacopter platforms, respectively, overlaid with fitted probability density functions (PDFs) of Normal, Gamma, Weibull, and Beta distributions. Among all candidate distributions fitted to the observed radial misalignment data for both platforms, the Gamma distribution demonstrated the best fit performance. For the quadcopter case, the Gamma model achieved the highest coefficient of determination, with $R^2 = 0.999$, indicating excellent agreement with the empirical PDF, while the hexacopter case exhibited similarly strong agreement with a slightly reduced but still high R^2 value.

B. Efficiency of Receiver Aperture Area

To evaluate the spatial efficiency of power capture, we examine the ratio of coupled power to receiver area, also known as the Expected Power-Area ratio, $\rho(A_r) = \mathbb{E}[P_r]/A_r$, across a range of receiver sizes. Fig. 8 illustrates the trade-off between receiver area and expected received power density for different receiver geometries under both stationary and moving conditions. Fig. 8(a) presents the logarithmic-scale power–area ratio for the quadcopter platform, revealing a distinct peak at small to moderate receiver areas, followed by a monotonic decline as the area increases. The log–log representation enables clear comparison across several orders of magnitude and highlights efficiency differences between aperture geometries. Fig. 8(b) shows the corresponding logarithmic-scale results for the hexacopter platform, exhibiting a similar trend but with the peak values around lower receiver area values due to reduced motion-induced beam spread. In all cases, moving receivers yield slightly lower peak power–area ratios compared to stationary ones as a result of beam smearing.

These results provide a practical design guideline: the receiver should be chosen as the smallest area that satisfies the required coupling, received-power, SNR or BER target under the expected UAV motion. Although larger apertures improve capture, they also increase size, weight, alignment burden, and potentially background noise. Therefore, for UAV-mounted FSO links, high-aspect-ratio apertures can be advantageous when the beam spread is anisotropic, since they improve coverage along the dominant displacement direction without increasing total receiver area. More broadly, because the ZOS model explicitly parameterizes link distance, beam divergence, atmospheric loss, turbulence strength, receiver geometry, and detector characteristics, the same framework can be used to perform parametric design sweeps and evaluate how UAV-FSO link performance scales beyond the fixed receiver-aperture cases studied here.

C. Communication Performance Metrics

The detector-level link performance is measured by first computing the optical power captured by the receiver aperture from the simulated beam footprint, followed by applying the receiver optics loss and a 20 dB attenuation stage before evaluating the resulting detector-input power against the operating limits of the Thorlabs PDA20C2 photodetector [24]. The detector has responsivity of 1 A/W at 1550 nm, 5 MHz bandwidth, NEP of 22 $\mathrm{pW}/\sqrt{\mathrm{Hz}}$, RMS noise of 11 mV and a 50 Ω transimpedance gain of 175 kV/A. Fig. 9(a) shows the received optical power at the aperture, after receiver optics loss, and after the attenuator, together with the minimum detectable power, the required 10 dB SNR threshold, and the detector saturation limit in the stationary receiver for the quadcopter case. Fig. 9(b) shows the corresponding detector SNR, where most samples remain above the 10 dB target, while sharp fading events reduce the SNR during severe pointing or atmospheric loss. Fig. 9(c) presents the estimated BER using the NEP-based SNR, showing that the link generally remains near very low BER values, while intermittent power drops produce short periods of degraded link quality.

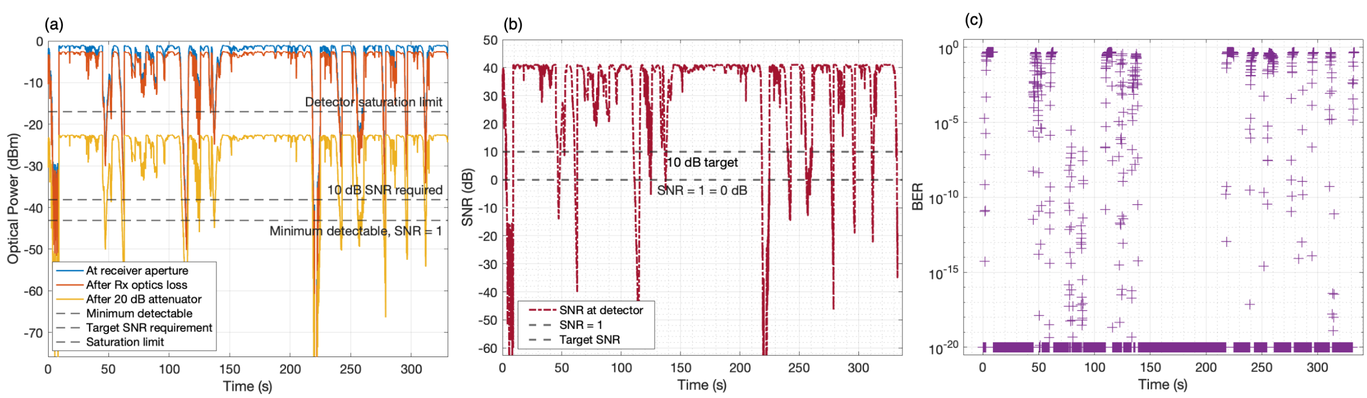

Fig. 9. Detector-level link performance showing received optical power budget, NEP-based SNR, and estimated BER after receiver optics loss and attenuation.

V. Optimizing Receiver Area

In an FSO communication system, the received optical power, P_r, at the detector is determined by several key factors: the transmitted power P_t, geometric beam spreading due to divergence over distance, atmospheric effects, optical system efficiency, and pointing stability affected by UAV vibration, deviation, or structural sway. Thus, the received optical power can be broadly expressed as

$$P_r = P_t \cdot \eta_{\text{geo}} \cdot \eta_{\text{atm}} \cdot \eta_{\text{sys}} \cdot \eta_{\text{point}}, \tag{1}$$

where η_{geo} denotes geometric attenuation due to beam divergence and receiver aperture size, η_{atm} represents the atmospheric channel efficiency, η_{sys} accounts for optical system transmission efficiency, and η_{point} models pointing loss due to misalignment or vibration.

A. Atmospheric Attenuation and Turbulence

In this work, the atmospheric channel efficiency is modeled as the product of two components: deterministic absorption loss and stochastic turbulence-induced fading. Therefore,

$$\eta_{\text{atm}} = \eta_{\text{abs}} \cdot \eta_{\text{turb}}, \tag{2}$$

where η_{abs} represents atmospheric absorption and $\eta_{\text{turb}}(t)$ represents normalized irradiance fluctuation due to atmospheric turbulence.

The absorption efficiency is modeled using the Beer–Lambert law as

$$\eta_{\text{abs}} = \exp(-\kappa_{\text{abs}} R), \tag{3}$$

where κ_{abs} is the atmospheric absorption coefficient and R is the propagation distance. If both absorption and scattering losses are considered, κ_{abs} can be replaced by the extinction coefficient, $\kappa_{\text{ext}} = \kappa_{\text{abs}} + \kappa_{\text{sca}}$.

Following the atmospheric turbulence model incorporated in [25] as a normalized stochastic fading coefficient,

$$\eta_{\text{turb}} = I_t, \qquad \mathbb{E}[I_t] = 1, \tag{4}$$

where I_t is the normalized irradiance fluctuation. Using the gamma–gamma turbulence model, the irradiance fluctuation is expressed as

$$I_t = I_x I_y, \tag{5}$$

where I_x and I_y represent the large-scale and small-scale turbulence effects, respectively. The probability density function of I_t is given by

$$f_{I_t}(I_t) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} I_t^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta I_t}\right), \qquad I_t > 0, \tag{6}$$

where $\Gamma(\cdot)$ is the gamma function, $K_{\alpha-\beta}(\cdot)$ is the modified Bessel function of the second kind, and α and β are the large-scale and small-scale turbulence parameters, respectively. These parameters are obtained from the Rytov variance as

$$\alpha = \left[\exp\left(\frac{0.49\sigma_R^2}{\left(1 + 1.11\sigma_R^{12/5}\right)^{7/6}}\right) - 1\right]^{-1}, \tag{7}$$

TABLE I
List of Symbols

Notation	Description
P_r	Received optical power at the detector
P_t	Transmitted optical power
A_r	Receiver aperture area
R_r	Receiver aperture radius, $R_r = \sqrt{A_r/\pi}$
D_l	Transmitter aperture diameter
θ_l	Beam half-angle divergence
R	Propagation distance
λ	Optical wavelength
k	Optical wave number, $k = 2\pi/\lambda$
θ	Pointing error angle, $\theta = r/R$
r	Radial beam-centroid displacement at receiver plane
r_{th}	Radial displacement threshold, $r_{\text{th}} = R\,\theta_l + \sqrt{A_r/\pi}$
w_z	Beam spot radius at receiver plane, $w_z = R\,\theta_l$
μ_x', μ_y'	Mean lateral displacements for deviation
σ_x', σ_y'	Standard deviations of vibrations in x and y axes
$f_r(r)$	Beckmann PDF of r
η_{geo}	Geometric attenuation, $\eta_{\text{geo}} = \dfrac{A_r}{\pi\big(\frac{D_l}{2} + R\,\theta_l\big)^2}$
η_{atm}	Atmospheric channel efficiency, $\eta_{\text{atm}} = \eta_{\text{abs}} \cdot \eta_{\text{turb}}$
η_{abs}	Atmospheric absorption efficiency, $\eta_{\text{abs}} = \exp(-\kappa_{\text{abs}} R)$
η_{turb}	Turbulence-induced fading factor, $\eta_{\text{turb}} = I_t$
I_t	Normalized irradiance fluctuation, $\mathbb{E}[I_t] = 1$
I_x, I_y	Large and small-scale turbulence irradiance components
κ_{abs}	Atmospheric absorption coefficient
κ_{sca}	Atmospheric scattering coefficient
κ_{ext}	Atmospheric extinction coefficient, $\kappa_{\text{ext}} = \kappa_{\text{abs}} + \kappa_{\text{sca}}$
C_n^2	Atmospheric refractive-index structure parameter
σ_R^2	Rytov variance, $\sigma_R^2 = 1.23 C_n^2 k^{7/6} R^{11/6}$
α, β	Gamma–gamma turbulence parameters
η_{sys}	Optical system transmission efficiency
η_{point}	Pointing-loss factor due to misalignment or vibration
$\mathcal{P}$	Outage probability, $\mathcal{P} = \Pr(r > r_{\text{th}})$
A_r^*	Optimal receiver aperture area

$$\beta = \left[\exp\left(\frac{0.51\sigma_R^2}{\left(1 + 0.69\sigma_R^{12/5}\right)^{5/6}}\right) - 1\right]^{-1}, \tag{8}$$

with the plane-wave Rytov variance given by

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} R^{11/6}, \tag{9}$$

where C_n^2 is the atmospheric refractive-index structure parameter and k is the optical wave number,

$$k = \frac{2\pi}{\lambda}. \tag{10}$$

Following the geometric attenuation model from [26], the received power can be written as

$$P_r = P_t \cdot \frac{A_r}{\pi\left(\frac{D_l}{2} + R\theta_l\right)^2} \cdot \eta_{\text{atm}} \cdot \eta_{\text{sys}} \cdot \eta_{\text{point}}, \tag{11}$$

where A_r is the receiver aperture area, D_l is the transmitter aperture diameter, θ_l is the half-angle divergence of the transmitter, and R is the propagation distance. Since $\eta_{\text{turb}}(t)$ is normalized, it does not change the mean received power but introduces instantaneous fading around the deterministic link-budget value.

Although the receiver plane in this work is analyzed in the y–z plane using UAV telemetry, the derivations in the

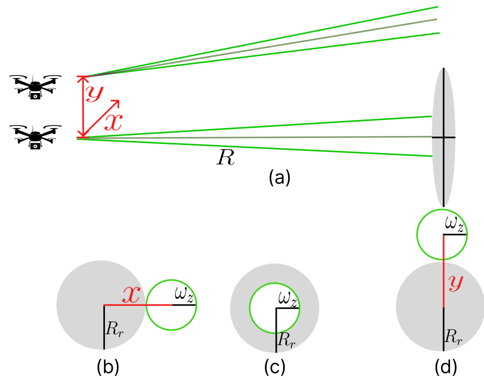

Fig. 10. (a) System schematic for FSO link between non-stationary transmitter and stationary receiver. Beam footprint on receiver aperture when there is (b) x-axis transmitter deviation , (c) perfect alignment, and (d) y-axis transmitter deviation.

following section are presented in the general x–y two-dimensional plane for clarity and ease of understanding.

B. Pointing Error, η_{point}

In real-world condition, especially with UAV-based FSO systems, pointing errors caused by vibration, deviation, or structural sway can lead to additional power loss, even when geometric alignment and atmospheric conditions are ideal. In [27], pointing error is modeled statistically as a Gaussian angular deviation, $\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, resulting in an exponential decay in received power due to misalignment. In addition, for UAV-based FSO links, both transmitter and receiver experience vibration-induced motion. The lateral displacement components at the receiver plane are modeled as [28]:

$$x_d = x_{\mathrm{r}} + R\theta_x, \quad y_d = y_{\mathrm{r}} + R\theta_y, \tag{12}$$

where x_{r} and y_{r} represent translational vibrations of the receiver in meters, and θ_x and θ_y are angular deviations in the x and y directions at the receiver. The resulting radial displacement of the beam centroid relative to the receiver aperture is then:

$$r = \sqrt{[x + (x_{\mathrm{r}} + R\theta_x)]^2 + [y + (y_{\mathrm{r}} + R\theta_y)]^2}. \tag{13}$$

This expression jointly accounts for transmitter vibration (x, y), receiver vibration $(x_{\mathrm{r}}, y_{\mathrm{r}})$, and angular deviation (θ_x, θ_y). These parameters are illustrated in Fig. 10.

The random displacements x and y are modeled as Gaussian random variables:

$$x \sim \mathcal{N}(\mu_x, \sigma_x^2), \qquad y \sim \mathcal{N}(\mu_y, \sigma_y^2), \tag{14}$$

where μ_x, μ_y capture static boresight offsets and σ_x^2, σ_y^2 denote beam vibration variances. Incorporating receiver vibration and angular deviation, the effective horizontal and vertical displacements become:

$$\tilde{x} = x + (x_{\mathrm{r}} + R\theta_x), \quad \tilde{y} = y + (y_{\mathrm{r}} + R\theta_y),$$

which are still Gaussian but with updated statistics:

$$\tilde{x} \sim \mathcal{N}(\mu'_x, \sigma'^2_x), \qquad \tilde{y} \sim \mathcal{N}(\mu'_y, \sigma'^2_y),$$

where

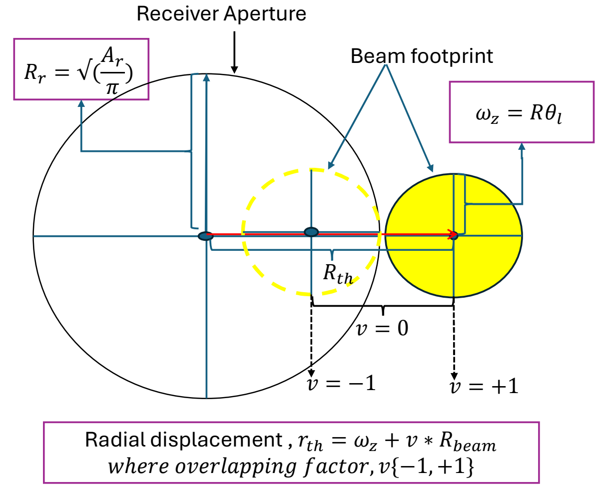

Fig. 11. System schematic for receiver aperture and beam footprint.

$$\mu'_x = \mu_x + \mathbb{E}[x_{\mathrm{tr}}] + R\ \mathbb{E}[\theta_{tx}], \tag{15}$$

$$\sigma'^2_x = \sigma_x^2 + \sigma_{x_{\mathrm{tr}}}^2 + (R\ \sigma_{\theta_{tx}})^2, \tag{16}$$

and μ'_y and σ'^2_y are modeled similarly. Thus, we model the pointing error η_{point} as the radial displacement $r = \sqrt{\tilde{x}^2 + \tilde{y}^2}$ which follows a Beckmann distribution [29]:

$$f_r(r) = \frac{r}{2\pi\sigma'_x\sigma'_y} \int_0^{2\pi} \exp\left[-\frac{(r\cos\phi - \mu'_x)^2}{2\sigma'^2_x} - \frac{(r\sin\phi - \mu'_y)^2}{2\sigma'^2_y} \right] d\phi, \quad r \geq 0. \tag{17}$$

This expression represents the exact distribution of the radial misalignment r for any arbitrary means and variances of the independent Gaussian-distributed x and y misalignment.

C. Outage Probability, $\mathcal{P}$

Consider the circular-shaped receiver of radius $R_r = \sqrt{A_r/\pi}$. The threshold radial displacement for outage can be formulated as

$$r_{\mathrm{th}} = w_z + R_r = v \cdot R\,\theta_l + \sqrt{\frac{A_r}{\pi}}, \tag{18}$$

where v represents the overlap factor, defined over the range $[-1, +1]$, which quantifies the degree of spatial overlap between the receiver aperture and beam footprint as shown in Fig. 11. When $v \leq 0$, the centroid remains inside the aperture and $v > 0$ the centroid moves outward. For any nonzero overlap, the beam centroid must satisfy $r \leq r_{\mathrm{th}}$. Deviations beyond this limit result in complete beam–aperture misalignment and hence link outage.

Given the outage probability $\mathcal{P}$, defined as the probability of complete misalignment (no overlap), is:

$$\mathcal{P} = \Pr(r > r_{\mathrm{th}}) = \int_{r_{\mathrm{th}}}^{\infty} f_r(r)\, dr. \tag{19}$$

Then, the probability of successful beam overlap is:

$$1 - \mathcal{P} = \Pr(r \leq r_{\mathrm{th}}) = \int_0^{r_{\mathrm{th}}} f_r(r)\, dr. \tag{20}$$

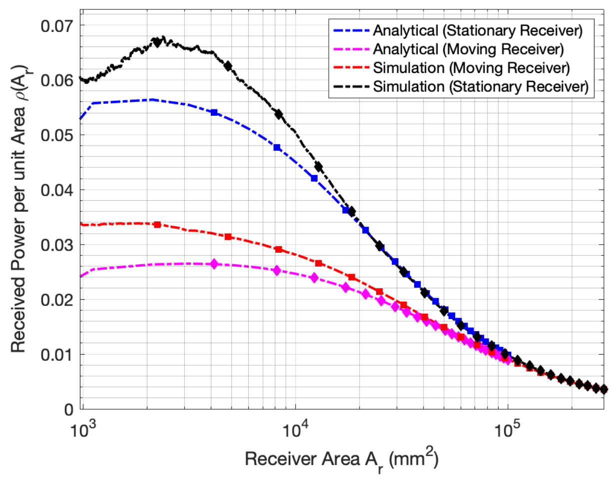

Fig. 12. Comparison between analytical predictions and simulation results for received power per unit area as a function of receiver area.

D. Expected Power-Area Ratio, $\rho(A_r)$

The received optical power, considering probabilistic beam misalignment, can be modeled as:

$$P_r(A_r) = (1 - \mathcal{P}) \cdot P_t \cdot \eta_{geo} \cdot \eta_{\text{atm}} \cdot \eta_{\text{sys}} + \mathcal{P} \cdot 0. \tag{21}$$

The first term on the right side represents the expected received power when the beam is successfully aligned with the receiver, accounting for geometric spread and pointing loss. The second term captures the outage case, where the beam entirely misses the receiver due to large pointing errors, resulting in zero received power. The expected received power is then:

$$\mathbb{E}[P_r(A_r)] = P_t \cdot \eta_{geo} \cdot \eta_{\text{atm}} \cdot \eta_{\text{sys}} \cdot \int_0^{r_{\text{th}}} f_r(r)\, dr. \tag{22}$$

which represents the average received power over the random angular deviations due to vibration. This expression provides a general analytical framework for computing the received power. The expected value of received power per unit area can be written as:

$$\begin{aligned}\rho(A_r) &= \mathbb{E}[P_r(A_r)]/A_r \\ &= \eta_{\text{atm}} \cdot \eta_{\text{sys}} \cdot \min\left(\frac{A_r}{\pi\left(\frac{D_l}{2} + R\theta_l\right)^2}, 1\right) \times \frac{P_t}{A_r} \\ &\int_0^{r_{th}} \int_0^{2\pi} \frac{r}{2\pi\sigma_x\sigma_y} \times \exp\left(-\frac{(r\cos\phi - \mu_x)^2}{2\sigma_x^2}\right. \\ &\qquad \left. - \frac{(r\sin\phi - \mu_y)^2}{2\sigma_y^2}\right) d\phi\, dr.\end{aligned} \tag{23}$$

In (23), the geometric attenuation factor $\eta_{\text{geo}}(A_r)$ is upper-bounded by 1, i.e.,

$$\eta_{\text{geo}}(A_r) = \min\left\{\frac{A_r}{\pi\left(\frac{D_l}{2} + R\theta_l\right)^2}, 1\right\}. \tag{24}$$

This bounding is necessary because the original linear expression for η_{geo} can exceed 1 for sufficiently large receiver areas, leading to an unphysical amplification of received power density $\rho(A_r)$. Since η_{geo} is intended to model the loss due to geometric spread, it should strictly remain within the interval $[0, 1]$. Allowing values beyond 1 artificially inflates $\rho(A_r)$ at large aperture sizes. By applying the `min` operator, we ensure that the geometric term remains bounded and interpretable as a normalized attenuation factor, preserving plausibility and preventing it from dominating the integral contribution in (23).

E. Receiver Power Density Model

Fig. 12 compares the analytical and simulation results for the received power per unit area $\rho(A_r)$ as a function of receiver aperture area A_r for both stationary and non-stationary receivers. The analytical curves are derived using the formulation in (23). In both cases, the beam was launched at a wavelength $\lambda = 1.31$ μm with a beam diameter $D_l = 15$ mm, propagating a distance of $R = 100$ m. This yields a divergence angle of $\theta_l = \lambda/(\pi D_l/2) \approx 5.56$ μrad and a total beam footprint radius at the receiver of $R_{\text{beam}} = D_l/2 + R\theta_l \approx 13.06$ mm. A beam truncation factor $r_{\text{th}} = v \cdot R\theta_l$ with $v = 0.1$ was applied as the upper bound of the radial integral in (23).

For the stationary case, the beam displacement is modeled as a bivariate normal distribution with zero mean ($\mu_x = \mu_y = 0$ mm) and standard deviations $\sigma_x = 72.46$ mm and $\sigma_y = 36.58$ mm. These values are obtained from Normal distribution fits to the histogram of beam center deviations for stationary receiver. The analytical and simulation curves show close agreement, especially across the mid-range aperture sizes, indicating that the reduced-order model captures the simulation trend under the extracted stationary beam-displacement statistics.

For the non-stationary (moving) receiver, the distribution is widened to reflect flight-induced jitter, with $\sigma_x = 102.38$ mm and $\sigma_y = 51.059$ mm, while keeping the mean centered at zero. These parameters are similarly extracted from Gaussian fits to beam displacement data recorded during UAV motion. As expected, the received power density drops more sharply compared to the stationary case due to increased beam spread and misalignment. Nonetheless, the analytical model retains excellent agreement with the simulation data, highlighting the robustness of (23) in modeling dynamic FSO links.

F. Optimization of Receiver Area for Maximum Power Density

To determine the most efficient receiver design, we optimize the receiver aperture area A_r that maximizes the received power per unit area $\rho(A_r)$ based on the analytical model presented in (16). The optimization problem is defined as:

$$A_r^\star = \arg\max_{A_r} \rho(A_r).$$

We perform numerical optimization using MATLAB's `fminbnd` solver, evaluating the integral at each step. We apply the optimization framework across a range of beam divergence angles θ_l to determine the optimal receiver area $A_r^\star$ that maximizes the received power density $\rho(A_r)$. As shown in Fig. 13, each divergence scenario exhibits a distinct $\rho(A_r)$ profile, where the received power density initially increases with area due to higher beam capture, then declines due to geometric attenuation. The location of the peak—marked by the circular and square markers—denotes the optimal $A_r^\star$.

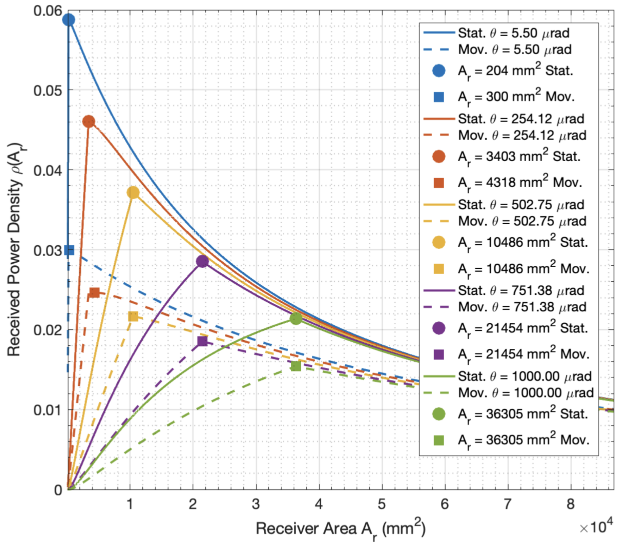

Fig. 13. Received power density $\rho(A_r)$ as a function of receiver area A_r for both stationary and moving receivers, across a range of divergence angles. Markers indicate the optimal area A_r^* that maximizes $\rho(A_r)$.

Motion-induced misalignment, modeled by a Gaussian beam center offset, leads to a slight shift in the optimal area, indicating the need for a larger receiver to compensate for dynamic fluctuations.

VI. Conclusion

In this work, we presented a measurement-informed simulation framework for evaluating UAV-induced misalignment in FSO communication links. UAV telemetry from controlled quadrotor hovering flights and publicly available hexacopter datasets was integrated into ZOS-based ray-tracing simulations to analyze beam displacement, received optical power, and coupling behavior under dynamic UAV motion. The results show that UAV vibration and motion significantly affect link alignment and received power, emphasizing the need for robust receiver design. Receiver aperture shape, size, and area were further evaluated to identify configurations that improve beam capture probability and maximize the expected power-area ratio. Future work will focus on real-time adaptive beam steering using gimbal- and MEMS-based correction, vibration-damping methods, and machine-learning-based motion prediction to improve alignment stability and FSO link performance.

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